

Article

Effectiveness of Floating Covers in Mitigating Ammonia and Hydrogen Sulfide Emissions from Lab-Scale Swine Slurry Pits

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Abstract: This study investigated the effectiveness of floating covers (FCs) in mitigating ammonia (NH₃) and hydrogen sulfide (H₂S) emissions from lab-scale swine slurry pits. Lab experiments were conducted over 125 days, comparing a treatment setup with FCs covering approximately 51.6% of the slurry surface to a control setup without covers. The results showed a significant reduction in NH₃ emissions by 54.4% with FCs ($p < 0.05$), which was attributed to their ability to limit NH₃ volatilization and promote crust formation. Although H₂S emissions were also reduced by 22.7%, this decrease was not statistically significant, likely due to the complex factors influencing H₂S production. These findings highlight the role of floating covers (FCs) in improving air quality within swine barns and reducing environmental pollution. By minimizing nitrogen loss as ammonia (NH₃), FCs enhance nitrogen recycling into agricultural land, supporting sustainable nutrient management. This aligns with broader sustainability goals by addressing air quality concerns, reducing odors, and improving resource efficiency in livestock systems. This study offers an effective method to mitigate air pollution, providing a foundation for practical and sustainable agricultural practices.



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Keywords: floating cover; ammonia; hydrogen sulfide; air pollution; environmental sustainability

1. Introduction

Ammonia (NH₃) emissions have become significant environmental concerns, particularly in the agricultural sector, where livestock production is a primary contributor. In 2021, the agricultural sector in OECD countries emitted approximately 13,601 thousand tons of NH₃, accounting for about 92.6% of the total NH₃ emissions [1]. Similarly, Korea reported approximately 262 thousand tons in the same year, with 76.5% (200 thousand tons) originating from agriculture [2]. Like OECD countries, Korea's agricultural sector thus represents the largest source of NH₃ emissions nationally. Within this context, the livestock industry alone contributed about 182 thousand tons, representing approximately 69% of the national total emissions and 91% of agricultural NH₃ emissions (Figure S1). The pig industry, in particular, was a significant contributor, accounting for 44% (79 thousand tons) of the livestock-related NH₃ emissions, highlighting the need for targeted strategies to mitigate this substantial environmental impact. NH₃ emissions from the pig farming

sector can be further classified into four categories based on emission sources: within swine barns, composting facilities, manure treatment facilities, and land application. Among these, emissions from within swine barns amounted to 25 thousand tons, accounting for 32.1% of the total pig industry emissions. Approximately 60% of pig housing facilities in Korea employ a slurry pit system, where manure is stored within the barn for extended periods [3]. The prolonged storage of manure in slurry pits releases harmful gasses, including NH_3 , hydrogen sulfide (H_2S), various odorous compounds, and greenhouse gasses. These emissions contribute to air pollution, acid rain, and climate change, posing significant risks to human health and the environment. NH_3 emissions, in particular, present serious environmental challenges; NH_3 acts as a precursor to fine particulate matter [4], exacerbates air pollution, and leads to soil acidification and eutrophication in aquatic ecosystems [5–7]. Although often overlooked, H_2S emissions also carry substantial environmental and public health risks. H_2S is highly toxic, with even low concentrations posing serious health threats, and prolonged exposure is linked to both acute and chronic health effects [8]. Additionally, H_2S contributes to air pollution, acid rain formation, and ecosystem degradation [9,10]. Therefore, effective management practices are crucial for mitigating these emissions and reducing the overall environmental impact of the pig farming sector. Considering the serious implications of both gasses, prioritizing the control of NH_3 and H_2S emissions is vital for environmental management and pollution prevention.

Various strategies have been implemented to reduce NH_3 emissions in pig farming operations. Low-protein feed formulations and phase feeding lower the nitrogen content in manure, thereby reducing NH_3 emissions during manure decomposition [11]. Additionally, rapidly removing manure from pits and implementing aerobically treated manure recharge systems minimize NH_3 emissions by reducing manure storage time in slurry pits [12,13]. Another effective strategy involves the use of additives. For instance, incorporating benzoic acid as a dietary additive or introducing sulfuric acid (H_2SO_4), hydrochloric acid (HCl), or superphosphate ($\text{Ca}(\text{H}_2\text{PO}_4)_2$) directly into the slurry effectively lowers the pH of pig manure [14–17]. This pH adjustment shifts the chemical equilibrium between ammonium (NH_4^+) and ammonia (NH_3), thereby significantly reducing ammonia emissions. In addition to these additive approaches, slurry pit cooling is another effective method for suppressing ammonia emissions, as lower slurry temperatures reduce ammonia emissions [18]. Among these strategies, technological interventions such as floating covers (FCs) have emerged as practical solutions to mitigate NH_3 emissions. By creating a physical barrier that limits gas exchange between the slurry surface and the atmosphere, FCs effectively reduce NH_3 emissions [19]. Floating covers are buoyant materials used to cover liquid surfaces and are commonly applied in lakes and reservoirs to reduce water evaporation in low-rainfall areas [20]. When used in livestock farming, FCs provide a practical and cost-effective solution for reducing odor and ammonia emissions from manure storage. Their efficiency, affordability, and adaptability make them suitable for use in manure storage tanks and directly over slurry pits.

Extensive research has been conducted on mitigating ammonia emissions during manure storage on livestock farms using various floating covers (Table S1). These studies, conducted by numerous researchers over the years, primarily focused on open manure storage systems, such as lagoons, ponds, and open slurry tanks. While reduction efficiencies varied depending on the specific cover type and its application, most studies reported substantial ammonia reductions, ranging from 50% to 95%. However, some studies indicated lower reductions, between 20% and 50%. A comprehensive review of these findings reveals that factors such as the type and thickness of the floating cover, the condition of the manure storage system, and seasonal variations significantly influence ammonia reduction efficiency. Despite significant ammonia and odor emissions from slurry pits in pig houses, research on applying floating

covers to these systems is limited, with only one study conducted by Santonja et al. [21]. Unlike open storage systems, slurry pits require covers that prevent manure accumulation on the surface due to the continuous fall of manure. Plastic films are unsuitable for slurry pits in this regard. Natural materials like straw, peat, and sawdust, while effective in preventing manure accumulation, tend to absorb moisture over time, leading to sinking. This can result in the blockage of manure transport pipes and increase organic matter content within the manure, thereby adding to the burden of subsequent treatment processes. To address these limitations, Santonja et al. [21] employed ball-type floating covers with a diameter of approximately 22 cm, covering about 70% of the slurry pit surface in a field-scale swine barn, and achieved a 28% reduction in ammonia emissions. In contrast, Loyon et al. [22] reported a 77% reduction when applying 20 cm diameter polystyrene balls to an open pig slurry storage container. This significant difference in results underscores the need for further research tailored to the unique environmental conditions of slurry pits. Matulaitis [23] conducted experiments using lab-scale cylindrical vessels (2.1 L volume) under enclosed conditions with controlled ventilation rates. However, their study focused solely on natural (peat, sunflower oil) and artificial (plastic film) floating covers, limiting the applicability of their findings to pig house slurry pits. To overcome these limitations, this study was primarily designed to evaluate the ammonia reduction efficiency of ball-type floating covers under conditions simulating the air exchange rate and temperature within actual slurry pits.

Previous studies (Table S1) have shown the inconsistent effects of floating covers on hydrogen sulfide reduction. For example, Matulaitis [23] reported reductions exceeding 74% for four out of six cover types, whereas peat and sunflower oil (under low-temperature conditions) increased hydrogen sulfide emissions. Applying floating covers to slurry surfaces is expected to reduce the exposed surface area, potentially creating more anaerobic conditions and increasing anaerobic odor compounds like hydrogen sulfide. However, Matulaitis [23] observed results contrary to this expectation. Therefore, this study also investigated the impact of ball-type floating covers on hydrogen sulfide emissions under controlled air exchange and temperature conditions.

This study aims to address the existing knowledge gap by conducting a laboratory-scale experiment designed to simulate real-world conditions within pig barns, focusing on the air exchange rate within the slurry pit headspace and the internal temperature of swine barns. By simulating these practical conditions, the findings provide valuable baseline information for scaling up floating cover technology to field-scale applications in swine barns, thereby supporting efforts to improve air quality and reduce odor and ammonia emissions in pig production systems.

2. Materials and Methods

2.1. Experimental Design

A lab-scale slurry pit was constructed using a rectangular polypropylene container with a capacity of 21 L (dimensions: 0.335 m length $\times$ 0.245 m width $\times$ 0.256 m height) to simulate a swine slurry pit, as shown in Figure 1. Two baffles were strategically positioned within the lab-scale slurry pit at regular intervals to enhance air contact across the slurry surface. This configuration enabled efficient airflow, entering from the inlet on one side of the pit, passing across the slurry surface, and exiting through the outlet on the opposite side, thus promoting effective air movement over the slurry.

The experiment consisted of a control group with an uncovered slurry pit and a treatment group with approximately 51.6% of the slurry pit surface covered by floating covers. Each group was tested in triplicate. The floating covers were 4 cm diameter plastic spheres. Spherical floating covers face challenges in achieving the complete coverage of the slurry surface due to gaps between the contact points of the balls. In the lab-scale slurry

pit used in this study, a maximum of 45 balls with a diameter of 4 cm could be placed. However, due to the density of the balls, the effective radius of the area in contact with the water surface was reduced to 1.73 cm, instead of the full 2 cm radius. Based on the surface area covered by the balls in contact with the liquid, the coverage rate was calculated to be 51.6% (Figure S2). This study utilized commercially available ABS (Acrylonitrile butadiene styrene) plastic balls as a floating cover for a lab-scale slurry pit evaluation of the ammonia mitigation potential of floating covers. While ABS plastic is known to pose potential environmental risks, such as microplastic generation and chemical leaching, under prolonged exposure to sunlight or harsh conditions, these risks were considered negligible in this short-term study conducted under controlled laboratory conditions without sunlight exposure. For future applications, polyethylene is the preferred material for floating balls due to its superior durability, resistance to degradation, and suitability for repeated use.

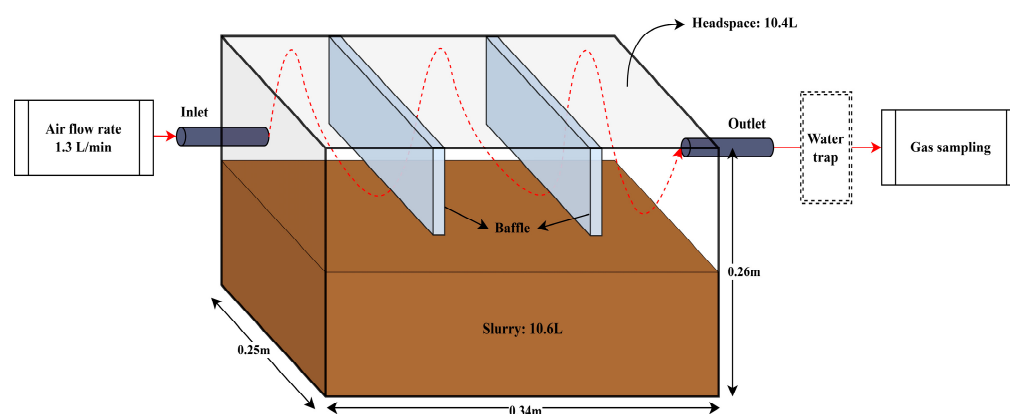


Figure 1. Schematic diagram of a lab-scale slurry pit system.

This experiment was designed to simulate the key environmental factors prevalent in pig barns. A total of 10.6 kg of swine slurry was added to each pit, maintaining a depth of approximately 13 cm, corresponding to about 50% of the pit's height. Ventilation was maintained at a flow rate of 1.3 L/min to replace the headspace volume above the pit approximately 7.5 times per hour, following the method outlined by Maurer et al. [24]. An airflow meter (LF-101, Unicell Instruments, Seoul, Republic of Korea) was installed between the pit's air inlet and the air pump (ACO-9730, Hailea, China) to ensure a consistent airflow rate. The experiment was conducted in a walk-in temperature-controlled chamber maintained at 25 °C to simulate the rearing temperature conditions of typical swine barns. T-type thermocouples, connected to a data logging system (CR1000 with AM16/32 relay multiplexer, Campbell Scientific, Inc., Logan, UT, USA), were used to monitor the temperature of the lab-scale slurry pits.

2.2. Characterization of Swine Slurry

Swine slurry was obtained from a finishing pig barn slurry pit and thoroughly mixed to ensure homogeneity. A representative sample was then collected. The manure was stored at 4 °C until use. Moisture content (MC) was determined by drying samples in an oven at 105 °C for 24 h. Volatile solids (VS) content was assessed by ashing the dried samples at 550 °C for 8 h in a muffle furnace. The pH was measured using an Orion 4 Star™ pH/conductivity benchtop meter (Thermo Fisher Scientific™, Waltham, MA, USA), while ammonium (NH_4^+) concentration was analyzed with a Gallery™ automated photometric analyzer (Thermo Fisher Scientific™, Waltham, MA, USA). Total nitrogen (TN) was analyzed using an Automatic Elemental Analyzer (Thermo Fisher Scientific™, Waltham, MA, USA).

2.3. NH_3 and H_2S Emissions Evaluation

NH_3 and H_2S concentrations were measured by collecting samples from the slurry pit outlet into a Tedlar[®] gas bag (Tedlar CEL Scientific, Santa Fe Springs, CA, USA). The gas samples were collected at a volume of 20 L per sampling event. NH_3 and H_2S concentrations were measured daily in the first week, once every two days in the second and third weeks, once every three days from weeks four to seven, and about once a week after the eighth week. This schedule allowed for comprehensive monitoring across all experimental phases. NH_3 concentrations in the collected air were determined using a tunable diode laser spectrometry-based laser gas detector (LGD Compact-A $\text{NH}_3/\text{H}_2\text{O}$, Axetris AG, Kaegiswil, Switzerland), while H_2S concentrations were measured with an electrochemical gas sensor ($\text{H}_2\text{S}/\text{C}-50$, Membrapor AG, Wallisellen, Switzerland). The emission rates of NH_3 and H_2S ($\text{g}/\text{day}/\text{m}^2$) were calculated using Equation (1).

$$G_E = (C_E - C_A) \times V \times \frac{MW}{22.141} \times \frac{273.15}{273.15 + T} \times \frac{1}{Ar} \times \frac{1}{1000} \quad (1)$$

G_E : Gas emission ($\text{g}/\text{day}/\text{m}^2$);

C_E : Gas concentration of exhausted air (mL/m^3);

C_A : Gas concentration of ambient air (mL/m^3);

V : Ventilation rate (m^3/day);

MW : Molecular weight of gas (g/mol);

T : Temperature ($^{\circ}\text{C}$);

Ar : Area of slurry pit simulation system (m^2).

2.4. Statistical Analysis

An independent samples t -test was conducted to evaluate statistically significant differences in NH_3 and H_2S emission rates from the slurry pits, comparing conditions with and without floating covers. The analysis was performed using OriginPro software (OriginLab, version 8.1), with a 95% confidence level.

3. Results and Discussion

3.1. Changes in Slurry Surface Crust Formation Within the Slurry Pit

Table 1 summarizes the visual changes observed on the slurry surface in the lab-scale slurry pits over the storage period. By day 28 of the experiment, the control group showed no surface crust formation, with only a few gas bubbles visible. In contrast, the treatment group exhibited an approximately 0.7 cm-thick, thin film forming on the slurry surface between the floating covers. By day 82, a barely perceptible thin crust (approximately 0.5 cm thick) was observed in the control group. A thicker crust of around 1.1 cm had formed in the treatment group. By day 125, the slurry in the control group rose to the surface, eroding the previously formed thin crust and leaving behind only a thin film. Meanwhile, in the treatment group, the crust had thickened to approximately 2.0 cm and had darkened noticeably.

Table 1. Variations in slurry surface crust development within slurry pit.

Day	Control		Treatment	
	Manure Surface	Observations on Crust Formation	Manure Surface	Observations on Crust Formation
0		Clean, no crust		Clean, no crust
	Crust Thickness: -		Crust Thickness: -	
28		Foam and initial crust formed		Thin, dry crust covered surface
	Crust Thickness: -		Crust Thickness: 0.7 cm	
82		Thin crust, nearly completely covered		Thick, dry crust covered surface
	Crust Thickness: 0.5 cm		Crust Thickness: 1.1 cm	
125		Crust disrupted, partially covered		Thick, dark, moist crust covered surface
	Crust Thickness: -		Crust Thickness: 2.0 cm	

During slurry storage, a natural crust layer often forms on the surface due to microbial biodegradation and evaporative losses. Crust formation and characteristics are influenced by factors such as solid content and volume-to-surface area ratio [25]. Slurry stored in tanks with low volume-to-surface area ratios typically forms crusts more rapidly, likely due to increased evaporative loss from the larger exposed surface area. In contrast, storage systems with higher volume-to-surface area ratios tend to produce thicker and stronger crusts over time, although their formation is slower because of reduced evaporative loss from the smaller exposed surface area [25].

In this study, the use of floating covers significantly influenced crust formation. The uncovered areas between the covers developed thicker and more stable crusts than the control group. This effect can be attributed to manure solids adhering to the undersides of the covers, which contributed to crust buildup in these areas (Figure S3). Furthermore, the floating covers provided physical protection to the developing crusts, preventing disruption and allowing consistent growth throughout the storage period.

3.2. Evaporation

Slurry evaporation over the 125-day experimental period was calculated by subtracting the combined weight loss due to sampling and volatile solids reduction from the total weight loss (Figure S4). Volatile solids in the slurry decreased over time due to the microbial decomposition of organic matter. Both the control and treatment groups showed a 42% reduction in volatile solids weight, corresponding to a loss of approximately 0.3–0.4 kg within the first 0.6 kg. The control group showed a water weight loss of 2.7 ± 0.2 kg, representing approximately 25.2% of the initial slurry input. In contrast, the treatment group, which utilized a floating cover, exhibited a water weight loss of 2.0 ± 0.1 kg, or about 19.4% of the initial slurry input. This indicates approximately 26% less evaporation in the treatment group compared to the control.

Several factors influence evaporation rates, including temperature, airflow, and surface area. Higher temperatures increase the kinetic energy of water molecules, facilitating their transition to the gaseous phase. Increased airflow removes saturated air from the liquid surface, promoting further moisture evaporation [20,26]. Using a floating cover can modify the microclimate above the slurry, reducing temperature fluctuations and limiting direct airflow over the slurry surface. This helps to suppress evaporation [27]. The observed 26% reduction in evaporation with a 51.6% surface coverage using floating covers demonstrates their effectiveness as a barrier to evaporation.

3.3. Slurry Characteristics

Table 2 presents the initial and final slurry properties. Over time, the moisture content (MC) of both the control and treatment groups slightly increased ($p < 0.05$). This increase in MC can be attributed to two factors: (1) the reduction in solid content due to organic matter decomposition and (2) water generation as a byproduct of microbial activity. Overall, the treatment group exhibited significantly higher MC than the control group ($p < 0.05$), primarily attributed to the floating cover's effectiveness in reducing evaporation.

Table 2. Variations in slurry characteristics after 125 days of storage in the slurry pit.

Characteristics	Control		Treatment	
	Initial	Final	Initial	Final
Moisture contents (w.b., %)	92.3 ± 0.2^a	93.9 ± 0.2^b	92.2 ± 0.03^a	94.9 ± 0.2^c
Volatile solids (d.b., %)	76.8 ± 0.2^a	62.5 ± 1.3^b	76.7 ± 0.2^a	62.4 ± 0.5^b
pH	6.8 ± 0.03^a	8.7 ± 0.07^b	6.8 ± 0.05^a	8.8 ± 0.03^b
NH ₄ -N (w.b., g/kg-slurry)	3.4 ± 0.1^a	2.2 ± 0.1^b	3.3 ± 0.03^a	3.2 ± 0.04^a
Total nitrogen (w.b., g/kg-slurry)	6.0 ± 0.1^a	4.2 ± 0.2^b	5.8 ± 0.2^a	4.8 ± 0.3^c

Values are means $\pm$ standard ($n = 3$) deviation. ^{a,b,c} Means with different superscript letters in the same row indicate a significant difference.

Volatile solids (VS) decreased in both the control and treatment groups over the storage period. The observed decrease in VS in this study can be attributed to the microbial degradation of organic material. This is consistent with previous studies showing that microorganisms decompose organic matter during manure storage, reducing volatile solids [28,29]. However, the use of floating covers did not significantly impact the reduction in VS in this study ($p > 0.05$).

The pH of the slurry increased significantly in both the control and treatment groups over the experimental period, rising from an initial value of 6.8 to 8.7 and 8.8, respectively ($p < 0.05$). This increase in pH can be attributed to the decomposition of organic nitrogen, which releases NH₃ during the prolonged storage period within the slurry pit. The NH₃ then ionizes in water to form NH₄⁺ ions, producing hydroxide ions that increase the pH of

the slurry. While both groups exhibited a significant pH increase, there was no significant difference in final pH between the control and treatment groups ($p > 0.05$).

Initially, there was no significant difference in total nitrogen (TN) concentrations between the control (6.0 ± 0.1 g/kg-slurry) and the treatment (5.8 ± 0.2 g/kg-slurry) groups ($p > 0.05$). However, by the end of the experiment, both groups exhibited a significant decrease in TN levels compared to their initial concentrations ($p < 0.05$). Additionally, there was a significant difference in TN concentrations between the control and treatment at the end of the experiment ($p < 0.05$). The control group had final TN concentrations of 4.2 ± 0.2 g/kg-slurry, while the treatment group had a final concentration of 4.8 ± 0.3 g/kg-slurry. At the start of the experiment, $\text{NH}_4\text{-N}$ accounted for approximately 57% of the TN in both the control and treatment groups. By the end of the experiment, this proportion increased to 66% in the treatment group but decreased to 52% in the control group. The $\text{NH}_4\text{-N}$ proportion in the control was approximately 14 percentage points lower than in the treatment, indicating that the absence of floating covers (FCs) in the control group led to significantly higher ammonia (NH_3) volatilization. This demonstrates the effectiveness of FCs in reducing NH_3 volatilization and minimizing nitrogen loss. The observed decrease in TN can be attributed primarily to nitrogen volatilization in the form of NH_3 and, to a much lesser extent, nitrous oxide (N_2O). However, as shown in Figure S5, N_2O emissions were negligible compared to NH_3 emissions and can be considered insignificant in this context. The smaller decrease in TN observed in the treatment group further highlights the role of FCs in mitigating nitrogen loss by limiting NH_3 volatilization.

3.4. NH_3 Emission

Figure S6 illustrates the NH_3 concentrations over the storage period. In the control group, NH_3 concentrations initially decreased to approximately 50 ppmv but subsequently increased steadily, peaking at around 560 ppmv between days 40 and 50. Following this peak, NH_3 levels gradually declined and stabilized at approximately 300–400 ppmv after 100 days of storage. These NH_3 concentration patterns align with the findings of Cortus et al. [30], who reported headspace NH_3 concentrations ranging from 250 to 820 ppmv in laboratory-scale slurry pits. In contrast, the treatment group maintained consistently lower NH_3 concentrations, ranging from 17 to 210 ppmv throughout the storage period. Over the entire 125-day storage duration, the average NH_3 concentration was 280.1 ± 3.5 ppmv in the control group, compared to 127.4 ± 3.2 ppmv in the treatment group. The control group exhibited NH_3 levels approximately 2.2 times higher than those observed in the treatment group, which was statistically significant ($p < 0.05$).

NH_3 emissions from manure pose risks to livestock productivity, workplace conditions, and human health [31]. High NH_3 levels can reduce feed intake in pigs, negatively impacting growth and reproductive performance. In humans, prolonged exposure to elevated NH_3 concentrations can lead to symptoms such as nausea, diarrhea, abdominal pain, mucosal irritation, and respiratory burns [32]. The substantial reduction in NH_3 concentrations observed with the use of floating covers in this study suggests that these covers may effectively mitigate these risks by significantly lowering NH_3 emissions.

Throughout the 125-day experimental period, the treatment group consistently exhibited lower NH_3 emission rates than the control group (Figure 2). During the early stage (0–27 days), both control and treatment groups showed increasing NH_3 emissions, likely due to the mineralization of organic nitrogen into NH_3 , a process driven by microbial activity and the conversion of NH_4^+ in slurry to NH_3 [33]. In the middle stage (28–81 days), the NH_3 emission trends began to diverge significantly between the control and treatment groups ($p < 0.001$). The control group experienced a sharp increase in NH_3 emissions from day 38, peaking at 8.8 g/day/m² on day 54, before declining by day 88. Conversely, the

treatment group maintained relatively stable NH_3 emission levels after the initial rise, with a maximum emission rate of 3.3 g/day/m^2 , which was 62% lower than the peak observed in the control group. This significant difference can be attributed to the larger exposed surface area in the control group, which facilitated more active aerobic ammonification at the slurry surface, in contrast to the treatment with floating covers that limited this exposure. After day 45, NH_3 emissions gradually declined in treatment groups, reaching a minimum of $1.0 \pm 0.1 \text{ g/day/m}^2$ on day 77. In the late stage (82–125 days), NH_3 emissions variability was more pronounced in the control group (± 0.0 – 3.5 g/day/m^2) than in the treatment group (± 0.0 – 0.7 g/day/m^2), suggesting that the floating covers not only reduced NH_3 emissions but also contributed to more stable emission levels over time. These results underscore the effectiveness of floating covers in reducing and stabilizing NH_3 emissions throughout extended storage periods.

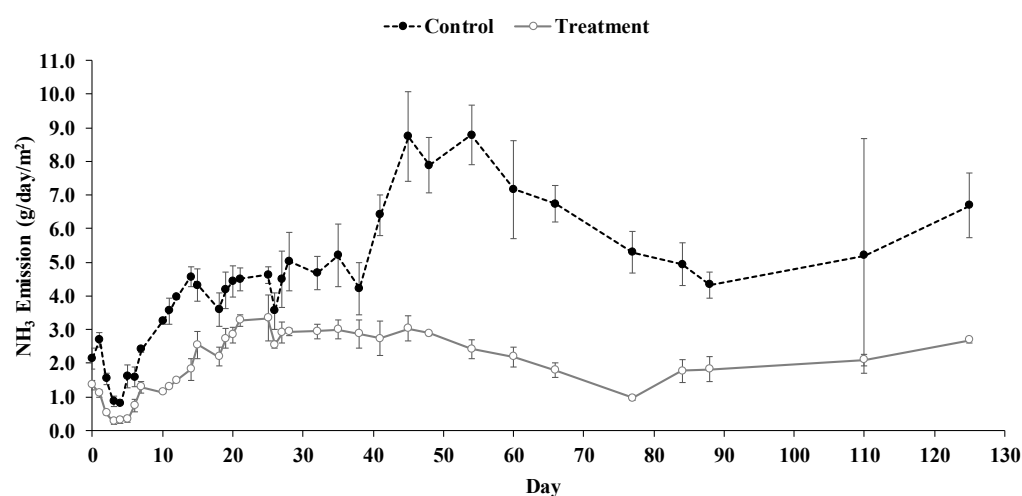


Figure 2. Variation of NH_3 emissions over the slurry storage period.

Table 3 summarizes the NH_3 emissions from the slurry pit during storage and the NH_3 emission reduction achieved with the floating cover. The average NH_3 emission rate was $4.4 \pm 0.1 \text{ g/day/m}^2$ in the control group and $2.0 \pm 0.0 \text{ g/day/m}^2$ in the treatment group, indicating an overall average reduction of 54.4% in NH_3 emissions due to the floating cover ($p < 0.001$). According to the study by Santonja et al. [21], a field-scale evaluation was conducted by placing 18 balls (225 mm in diameter) per square meter in a slurry pit, covering 70% of the emitting surface area. However, they observed only a 28% reduction in ammonia emissions. In contrast, this study demonstrates an approximately twofold higher reduction efficiency. It is necessary to examine whether this difference is due to the disparity between lab-scale and field-scale conditions or other influencing factors. NH_3 emissions within pig barns occur as NH_3 volatilizes from the slurry surface and diffuses into the atmosphere [34]. The NH_3 volatilization rate is influenced by factors such as the NH_3 mass transfer coefficient and the partial pressure gradient between the boundary layer and the surrounding air. Key determinants of the NH_3 mass transfer coefficient include temperature, wind speed across the boundary layer, and the exposed emission surface area [35]. The observed reduction in NH_3 emissions when using the floating cover can be attributed to its effectiveness in decreasing the exposed surface area of the slurry, thereby limiting NH_3 volatilization.

As shown in Table 3, the NH_3 reduction efficiency of the floating cover was approximately 45.7% during the initial storage phase (0–27 days). In the middle (28–81 days) and late (82–125 days) stages, this efficiency increased to around 60.4%, suggesting that the floating cover's effectiveness in reducing NH_3 emissions improved over time. This

improvement may be due to increased NH_3 production as microbial decomposition of organic nitrogen intensifies with prolonged storage. When NH_3 generation is high, the floating cover's impact on reducing NH_3 volatilization becomes more pronounced. Over time, the NH_3 volatilization rate in the control group significantly increased, doubling during the middle stage and rising by 1.7 times in the late stage compared to the initial stage. In general, higher temperatures and alkaline pH conditions promote the conversion of ionized ammonia (NH_4^+) to free ammonia (NH_3), which is more volatile [36]. However, in this study, the temperature was fixed at 25 °C, eliminating the influence of temperature fluctuations on NH_3 volatilization. Regarding pH, the initial value of 6.8 increased significantly to 8.7 in the treatment and 8.8 in the control by the end of the storage period. This increase in pH had a substantial impact on ammonia volatilization, as higher pH levels favor the formation of NH_3 over NH_4^+ , thereby enhancing the volatilization process [37]. According to Emerson, Russo, Lund, and Thurston [36], at 25 °C and pH 7, only 0.566% of NH_3 exists as NH_3 , whereas at pH 9, this proportion rises to 36.3%. Therefore, the observed increase in NH_3 volatilization with storage time can be attributed to the accelerated microbial decomposition of organic nitrogen and the rise in pH.

Table 3. Comparison of NH_3 emissions and reduction rates across slurry storage times.

Day	NH_3 Emission (g/day/m ²)			Reduction Rate (%) ²
	Control	Treatment	<i>p</i> -Value ¹	
0–27 (<i>n</i> = 60)	3.1 ± 0.2 ^a	1.7 ± 0.1 ^a	<0.001	45.7
28–81 (<i>n</i> = 33)	6.4 ± 0.4 ^b	2.5 ± 0.1 ^b	<0.001	60.4
82–125 (<i>n</i> = 12)	5.3 ± 0.8 ^b	2.1 ± 0.2 ^c	0.002	60.4
Overall (<i>n</i> = 105)	4.4 ± 0.1	2.0 ± 0.0	<0.001	54.4

Values are means ± standard deviation. ¹ The *p*-value represents a comparison of the means between the control and treatment within the same row. ² This presents a percentage reduction compared to the control. ^{a,b,c} Means with different superscript letters in the same column indicate a significant difference.

3.5. H_2S Emission

Figure S7 illustrates the temporal variation in H_2S concentrations across the slurry storage period. In the control group, H_2S levels ranged from 64 to 17,370 ppbv, while in the treatment group, they varied from 53.5 to 25,152 ppbv. The overall mean H_2S concentration measured over the 125-day experimental period was 6292 ± 1611 ppbv in the control group and 4866 ± 891 ppbv in the treatment group. Although the control group showed H_2S concentrations approximately 1.3 times higher than the treatment group, this difference was not statistically significant ($p > 0.05$).

H_2S is produced primarily under anaerobic conditions by the activity of sulfate-reducing bacteria and also by the decomposition of sulfur-containing amino acids such as cysteine and methionine [38]. pH levels significantly influence the dissociation and release of H_2S generated under anaerobic conditions within slurry. At lower pH, H_2S exists primarily in its undissociated form (H_2S), which has low solubility in water. This leads to supersaturation, resulting in the formation of microbubbles that rapidly burst [39], releasing a significant amount of H_2S trapped within [40]. The high initial H_2S concentration observed in both the control and treatment groups is likely due to the low initial pH of 6.8. In contrast to the initial concentrations, H_2S concentrations were very low in both groups at later storage stages. Once the pH exceeds 7, H_2S breaks down into HS^- and S^{2-} ions. The increased solubility of these ions at higher pH values likely explains the lower H_2S concentration observed at later storage stages. The final pH values of 8.7 and 8.8 for the control and treatment groups, respectively, support this explanation.

Figure 3 illustrates the H_2S emission rates over the slurry storage period. The average H_2S emission rate was 197.4 ± 50.8 mg/day/m² in the control group and

152.5 ± 28.0 mg/day/m² in the treatment group, representing an approximately 22.7% reduction in H₂S emissions due to the use of the floating cover. Despite this observed reduction, the difference was not statistically significant ($p > 0.05$).

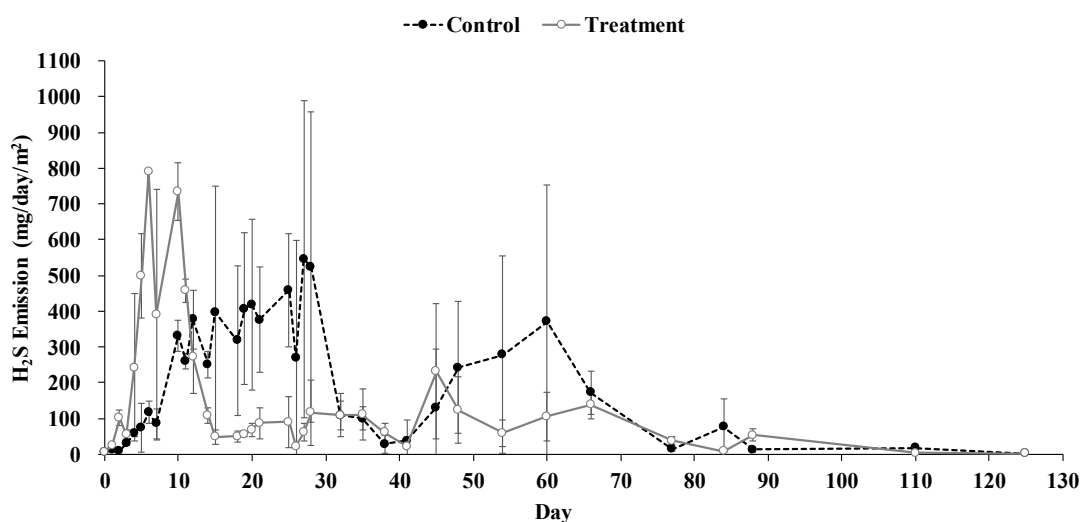


Figure 3. Variation of H₂S emissions over the slurry storage period.

As shown in Table 4, the reduction rates of H₂S emissions in the treatment group compared to the control were 13.3%, 44.3%, and 38.9% during the early, middle, and late stages of the experiment, respectively. However, none of these reductions were statistically significant ($p > 0.05$). Despite the lower average H₂S emissions observed in the treatment, high variability across replicates likely contributed to the lack of statistical significance, suggesting that the floating cover's effect on H₂S emissions was inconsistent. This variability may indicate that factors beyond the application of the floating cover influenced the outcomes. Several external factors impact H₂S emissions during manure storage, including temperature, pH, and moisture content, which influence the anaerobic environment that is essential for the sulfate-reducing bacteria responsible for H₂S production [41]. The release patterns of H₂S from swine slurry are complex, irregular, and challenging to predict, indicating a need for further research to fully understand the factors influencing H₂S emissions [42].

Table 4. Comparison of H₂S emissions and reduction rates across slurry storage times.

Day	H ₂ S Emission (mg/day/m ²)			Reduction Rate (%) ²
	Control	Treatment	<i>p</i> -Value ¹	
0–27 (<i>n</i> = 60)	239.9 ± 53.1 ^a	207.9 ± 18.7 ^a	0.38	13.3
28–81 (<i>n</i> = 33)	182.2 ± 60.8 ^a	101.4 ± 55.3 ^b	0.16	44.3
82–125 (<i>n</i> = 12)	26.6 ± 18.8 ^b	16.3 ± 4.7 ^c	0.41	38.9
Overall (<i>n</i> = 105)	197.4 ± 50.8	152.5 ± 28.0	0.25	22.7

Values are means ± standard deviation. ¹ The *p*-value represents a comparison of the means between the control and treatment within the same row. ² This presents a percentage reduction compared to the control. ^{a,b,c} Means with different superscript letters in the same column indicate a significant difference.

Unlike NH₃ emissions, which increased during the middle and late stages, H₂S emissions were highest during the early stage for both the control and treatment groups and gradually declined as storage time progressed. pH plays a significant role in H₂S emissions by affecting its ionization in solution [43]. As pH increases, H₂S dissociates into HS[−] and S^{2−} ions, reducing gaseous H₂S emissions [44]. Ref. [45] observed that at 25 °C and pH 7.0, around 50% of H₂S remains in its undissociated form, while at

pH 9.0, this proportion decreases to approximately 1%. In this study, the pH increased from 6.8 to 8.7 in the control group and 8.8 in the treatment group over the experimental period. Consequently, H₂S emissions decreased significantly, by 88.9% in the control group and 92.2% in the treatment group, respectively. These results align with the theoretical relationship between pH and H₂S emissions, suggesting that the observed reduction was primarily due to the rise in pH over time.

In the control group, the difference in average H₂S emissions between the early and middle stages was not statistically significant ($p > 0.05$); however, emissions decreased significantly by the late stage ($p < 0.05$). In contrast, the treatment group exhibited a significant reduction in H₂S emissions from the early to the middle and late stages ($p < 0.05$), indicating that the floating cover may facilitate a faster reduction in H₂S emissions over time. As shown in Table S2, the treatment group exhibited a more rapid pH increase than the control group. This accelerated pH rise likely contributed to the greater reduction in H₂S emissions observed in the treatment group.

4. Conclusions

While previous studies primarily focused on applying floating covers (FCs) to open manure storage facilities, this study uniquely investigates the potential of ball-type floating covers for swine barn slurry pits, highlighting their distinct advantages over existing approaches. Specifically, this study demonstrates the significant potential of ball-type FCs to mitigate NH₃ emissions from swine barn slurry pits. Lab-scale slurry pit experiments demonstrated that ball-type FCs, covering 51.6% of the slurry surface, significantly decreased NH₃ emissions by approximately 54% over 125 days. This reduction was attributed to the FCs acting as a physical barrier, which limits NH₃ volatilization and promotes crust formation on the slurry surface. Generally, a higher coverage rate is expected to lead to greater ammonia emission reduction. However, achieving complete coverage with spherical floating covers is challenging due to gaps between the contact points of the balls. To address this issue, increasing the density of ball-type floating covers in field applications could improve the coverage rate from approximately 52% to around 69%. This increase in coverage is also anticipated to enhance ammonia reduction efficiency. Although the impact on H₂S emissions was less significant, the findings demonstrate the potential of FCs as a promising strategy for reducing NH₃ emissions in swine production systems. However, lab-scale experiments cannot fully replicate real-world swine barns' dynamic and complex conditions. Factors such as manure depth, slurry accumulation rates, pig stocking densities, and varying climate conditions can significantly affect the performance of FCs in practical applications. Additional challenges, including continuous manure deposition, feed and water spillage, and seasonal variations in temperature and humidity, must be addressed to improve the technology for field use. Addressing these complexities is essential for advancing FC technology into a sustainable and practical solution for reducing NH₃ emissions in swine barns. By minimizing nitrogen losses through reduced NH₃ volatilization, FCs can enhance nitrogen use efficiency and enable the recycling of nitrogen resources back into agricultural land. Such advancements will improve air quality, reduce the environmental impact of swine production, and contribute to broader goals of agricultural sustainability and environmental protection.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/su17010374/s1>, Figure S1: Annual ammonia emissions in Korea in 2021 (NAIR, 2021); Figure S2: Surface area coverage rate by floating cover; Figure S3: Crust adhering to the underside of the floating cover (Day 125); Figure S4: Water evaporation mass balance results; Figure S5: The change of NH₃-N, N₂O-N over the slurry storage period; Figure S6: Variation of NH₃ concentration over the slurry storage period; Figure S7: Variation of H₂S concentration over

the slurry storage period; Table S1: Previously studied floating cover types, applied facility categories, and ammonia and hydrogen sulfide reduction efficiencies; Table S2: Comparison of pH changes over slurry storage periods [12,19,22,23,26,46–55].

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